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Acute effects of overspeed stimuli with towing system on athletic sprint performance: A systematic review with meta-analysis

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ABSTRACT

Overspeed-based training is widely used to improve athletes' maximum running speed and towing systems are one of the most frequently employed methods for this purpose. However, the effectiveness of this modality has not been thoroughly determined. This review analyzes the acute effects of overspeed conditions with towing systems in sprinters. The articles were searched, analysed and selected following the PRISMA methodology in the PubMed, SPORTDiscus and Google Scholar databases. Sixteen studies were included, with a total sample of 240 men and 56 women (14 to 31y; 1.73 to 1.82 m; 66.2 to 77.0 kg). The main acute responses found were: 1) an increase in maximum running speed (ES = 1.54, large), stride length (ES = 0.92, moderate), flight time (ES = 0.28, small) and stride rate (ES = 0.12, trivial); and, 2) a decrease in contact time (ES = 0.57, small). However, analysis of the reported ground reaction forces and electromyography data did not provide enough consistent evidence to conclusively determine whether the changes are due to a greater muscular response of the athlete or the effect of the towing system. Future research should focus on studying the mechanisms responsible for the observed acute effects.

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KEYWORDS
Assisted sprint; supramaximal speed; overspeed overload; individualization

Introduction

Maximal running speed (MRS) is a critical determinant of performance in athletics speed disciplines (K Clark et al., 2021; Haugen et al., 2019; Leblanc & Gervais, 2004; Mero et al., 1992; Slawinski et al., 2017). Training protocols to improve speed performance are based on the principle of specificity. These may include the application of assisted and resisted forces during running (Faccioni, 1994; Hicks, 2017; Kristensen et al., 2006). The assisted and resisted method for speed implies a greater neuromuscular stimulus. However, the use of both methods must be carried out carefully in order not to impair running mechanics (Cissik, 2005; K Clark et al., 2021; Faccioni, 1994; Hauschildt, 2010; Leyva et al., 2017).

Assisted speed methods, known as supramaximal speed or overspeed (OS), allow athletes to reach higher speeds than their MRS (K Clark et al., 2021). These kinds of stimuli have been traditionally employed by sprinting coaches mainly because of a specific overload principle. This is achieved using a higher speed than the speed usually reached by the athlete without assistance (Van Den Tillaar, 2004). One important concept for speed athletes is the term "speed barrier", which refers to the limit of velocity that an athlete could reach (Sedlácek et al., 2004, 2015; Van Den Tillaar & Gamble, 2017, 2018). Hypothetically, OS training could help exceed this speed barrier and improve sprint performance acutely or in the long term when the stimulus is applied in a consistent and progressive manner (Cissik, 2005; Sedlácek et al., 2015).

With OS, the increased requirements on the high nerve conduction velocity demand could cause neural adaptive responses, particularly in the best sprinters since they showed an increased proportion of active fast-twitch muscle volume (Mero & Komi, 1985). Therefore, OS training would be a functional way for sprinters to develop specific strength and increase MRS, producing a higher physiological stimulus on the neuromuscular system in comparison with normal sprint training (A Ross et al., 2001; RE Ross et al., 2009). These changes produced in the nervous system by OS could adapt neuromuscular performance to a higher level (Hopkins et al., 2009).

OS training could be accomplished with different strategies such as downhill running (Ebben, 2008; Paradisis & Cooke, 2001), high-speed treadmill (Hauschildt, 2010) or using a towing system (TS). The TS consists of providing the athletes with external assistance that allows them to run faster than their MRS while being pulled in the direction they are running (K Clark et al., 2021; DA Clark et al., 2009; Leblanc & Gervais, 2004). This assistance can be provided by different means: a) by another person attached to a system of rubber bands or cords and pulleys that multiply forces (Corn & Knudson, 2003;

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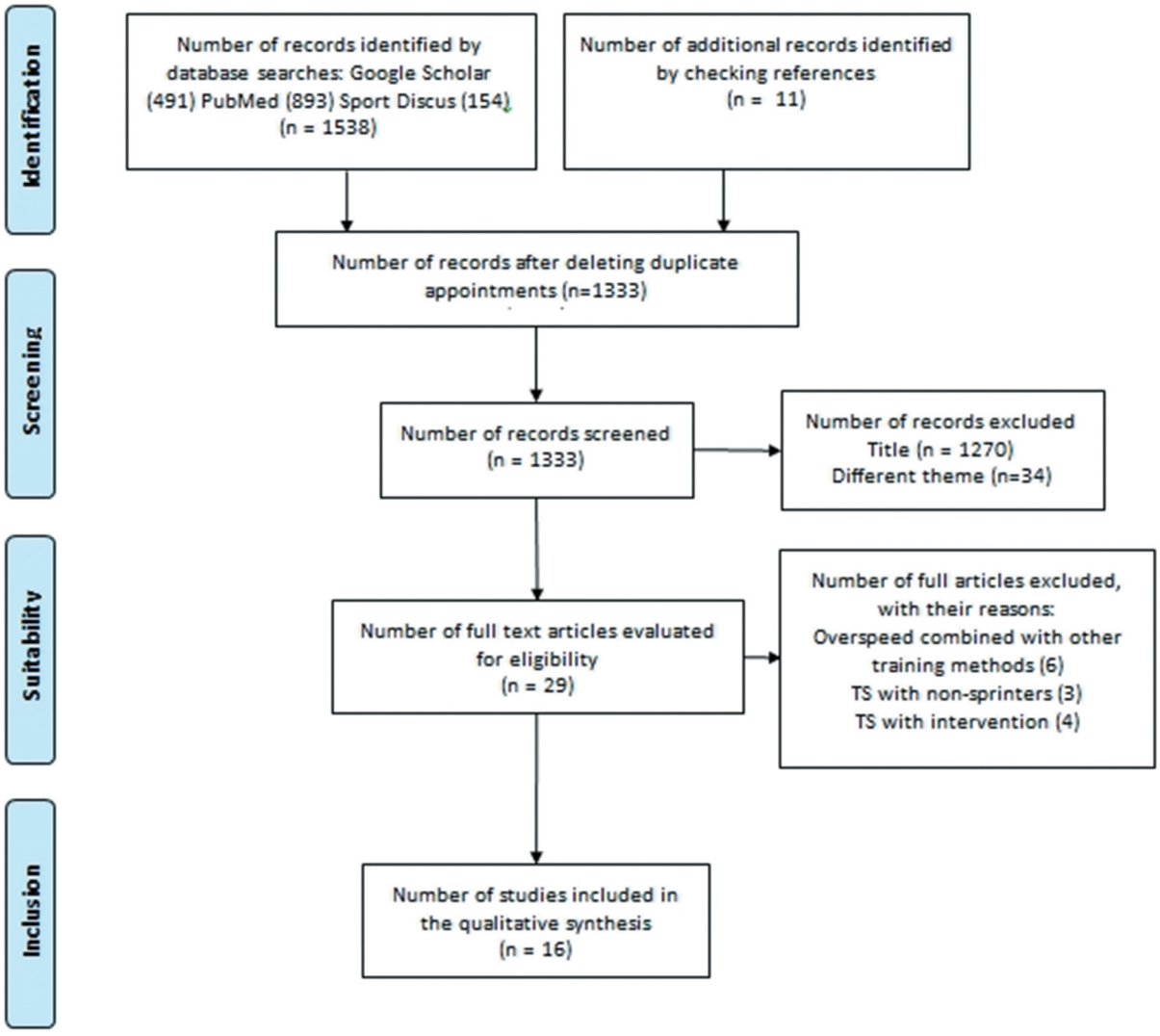

Figure 1. PRISMA Flow chart. PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analysis. Moher et al. (Mero & Komi, 1987); TS: towing system.

Leblanc & Gervais, 2004; Mero & Komi, 1985, 1986, 1987; Sedlácek et al., 2004, 2015); b) by a rubber band or counter-weight systems and pulleys with no participation of another person (Colyer, Nagahara, Salo et al., 2018; Stoyanov, 2019; Sugiura & Aoki, 2008) and; c) by motorized systems (Bosco & Vittori, 1986; K Clark et al., 2021; Mero et al., 1987; Sugiura & Aoki, 2008; Van Den Tillaar, 2018; Van Den Tillaar, 2021). Motorized systems are usually controlled by a computer and consist on dragging the runner in the running direction using a cable and a motor. These forces can be adjusted more or less precisely depending on the device chosen. Normally, they are applied in a constant (isotonic) way to the runner, both during the flight phase and the contact phase (K Clark et al., 2021; Sugiura & Aoki, 2008). The technical specifications of each TS as well as descriptive figures can be consulted in the studies included in this paper.

The literature states that exposure to OS stimuli may promote acute effects (Bosco & Vittori, 1986; K Clark et al., 2021; DA Clark et al., 2009; Corn & Knudson, 2003; Leblanc & Gervais, 2004; Mero et al., 1992; Sedlácek et al., 2015; Sugiura & Aoki, 2008; Van Den Tillaar, 2018; Van Den Tillaar, 2021) and may improve sprint performance in the long-term (Kristensen et al., 2006; Majdell & Alexander, 1991; Rumpf et al., 2016; Stoyanov, 2019). However, there is scant discussion as to which biomechanical, neuromuscular and physiological changes are encouraged by this type of training (Kristensen et al., 2006; Leblanc & Gervais, 2004). Moreover, it hasn't been possible to determine exactly if the main acute changes observed at the kinematic or biomechanical levels are due to greater muscle activation (Mero & Komi, 1986, 1987; Mero et al., 1987; Sugiura & Aoki, 2008; Van Den Tillaar & Gamble, 2018), to an increase in ground reaction forces (Mero & Komi, 1986, 1987; Mero et al., 1987) or to the action of the system that drags the athlete forward (Leblanc & Gervais, 2004; Sedlácek et al., 2015; Sugiura & Aoki, 2008).

The main aim of this review was to analyse the TS method among all OS methods, taking into account that it allows greater control in the quantification of the external traction load and that the stimulus may be adjusted individually for each athlete (K Clark et al., 2021; Corn & Knudson, 2003; Sugiura & Aoki, 2008; Van Den Tillaar, 2018). In this regard, this review sought to establish methodological recommendations for

possible OS with TS interventions. These recommendations may influence: a) the intensity of the OS with TS that allows the MRS to be exceeded without altering technical and biomechanical patterns, with an impact on load customization; b) the use of TS systems that allow the OS conditions to be replicated in an exact manner and to obtain immediate performance feedback.

Methods

Search strategy and data sources

The first search was conducted in June 2018 while the last search was carried out in August 2021. The following databases were used: PubMed, Google Scholar, and SPORTDiscus. Following the guidelines of each database, the search strategy used was: ([athletes OR players OR runners] AND [superspeed OR overspeed OR "supramaximal speed" OR "wind-assisted sprint" OR "assisted sprint" OR "elastic cord" OR downhill OR treadmill OR "towing sprint"] AND [acceleration OR "maximal sprint speed" OR "maximal running velocity"]). Additionally, a manual search was performed using the bibliographies of the included articles with the aim of identifying additional studies.

Inclusion and exclusion criteria

This review included articles published in scientific journals in addition to congress and conference documents obtained from the aforementioned databases, with no date limits. Only studies developed with healthy individuals were taken into account. Only articles that met the following inclusion criteria were considered: 1) studies developed with sprinters; 2) studies with data on the effects of acute exposure to OS conditions with TS; 3) studies with data on the TS system and loads used; 4) studies with standardized measures of interest, mainly MRS, contact time (CT), flight time (FT), stride length (SL), and stride rate (SR).

Data extraction

The results of the database search were imported into the bibliographic management software (Mendeley Desktop version 1.19.4 for Windows) and duplicate copies were removed. Then, a rapid assessment was performed to analyse and discard articles based on titles and abstracts. Subsequently, articles with the potential to be included were read in full to determine whether they were suitable for inclusion in the review or not. The selection process applied to the articles screened was based on the selection of population types, types of intervention methods and types of variables analysed.

The studies included in this review focused on the relationship between OS and the following variables: MRS, SL, SR, CT, FT, running or technical ability skills, data of electromyographic activity, ground reaction forces, spring constants or measures of stiffness and metabolic data. Furthermore, the following relevant aspects were included to analyse the studies: title, authors, name and type of publication, sample size and characteristics, sample experimental protocol, training load, parameters and typology of the TS, results of each parameter, and between-group differences in the study that may be relevant for this review.

Statistical analysis

Jamovi for Mac (version 1.6.21; The Jamovi project (2021), Sidney, Australia, retrieved from https://www.jamovi.org) and Microsoft Excel for Mac (Microsoft Office 16.47.1; Microsoft Corporation, 2021) were used to carry out the meta-analysis.

A random-effects meta-analysis (Hagger, 2006) was performed to determine the overall effect of overspeed conditions for the different kinematic outcomes. The kinematic outcomes considered were: velocity increases (**Δ** vel), SL, SR, CT of the eccentric phase, CT of the concentric phase, CT of the concentric and eccentric phase and FT.

As all the included studies had within-subjects designs, the differences between overspeed conditions and maximal sprint conditions were expressed as effect sizes (ES) using the standardized mean change (Botella Ausina & Sánchez-Meca, 2015; Morris, 2000) with 95% confidence intervals (CI). Effect sizes were interpreted as: < 0.2 = *trivial*; 0.2–0.6 = *small*; 0.6–1.2 = *moderate*; 1.2–2.0 = *large*; > 2.0 = *very large* (Hopkins et al., 2009).

Heterogeneity between studies was evaluated with I^2 statistics and between-study variance with the tau-square (τ^2) (Borenstein et al., 2009). The magnitude of heterogeneity for results was classified according to the following scale: < 25% (low), 25–75% (medium), and > 75% (high) (Borenstein et al., 2009). If the *p* value for the $x^2 < 0.1$, this indicated the presence of heterogeneity, with a τ^2 value > 1 suggesting the presence of substantial statistical heterogeneity (Borenstein et al., 2009).

Risk of bias assessment

To analyse the risk of bias, the studies included were assessed by two independent raters (PCG and JVBG), while disagreements were resolved by consultation with a third party (JAH). The "Risk of Bias in Non-Randomized Studies – of Interventions (ROBINS-I)" tool was used (Sterne et al., 2016). The ROBINS-I includes signalling questions alongside free-text boxes within each of the seven domains of bias to facilitate the evaluations of the risk of bias. The first two domains address issues before the start of the interventions that are to be compared. The third domain addresses issues during the interventions. The other four domains address issues after the start of interventions. Each domain was evaluated as: 1 – low risk of bias; 2 – moderate risk of bias; 3 – serious risk of bias; 4 – critical risk of bias; and, 5 – no information). The evaluations of the risks of bias were summarized across all studies included for each of the domains listed. The risk of bias assessment was plotted in Figure 3 using the *Robvis tool* (McGuinness & Higgins, 2021).

Results

Search, screening and selection of results

1,538 articles were found in the databases. In addition, a further 11 were added from the reference lists. Finally, 16 articles were included in this review. The complete process of article analysis,

Table 1. Studies included in this systematic review: type and sample characteristics.

Reference	Type	n	Males	Females	Groups	Age (y)	Height (m)	Weight (kg)
(Mero & Komi, 1985)	Journal	22	13	9	4	20.4	1.75	67.3
(Bosco & Vittori, 1986)	Journal	4	4	0	1	21.5 ± 2.1	1.77 ± 0.02	69.0 ± 7.0
(Mero & Komi, 1986)	Journal	19	11	8	3	20.9	1.77	68.9
(Mero et al., 1987)	Journal	9	9	0	1	25.5 ± 2.2	1.81 ± 0.05	73.7 ± 4.2
(Mero & Komi, 1987)	Journal	19	11	8	3	20.9	1.77	68.9
(Corn & Knudson, 2003)	Journal	9	5	4	1	21.2	1.73	68.5
(Leblanc & Gervais, 2004)	Conference	6	5	1	1	21.8 ± 1.7	1.78 ± 0.08	77.0 ± 8.6
(Sedlácek et al., 2004)	Journal	52	52	0	2	14 to 31	N/D	N/D
(Sugiura & Aoki, 2008)	Journal	8	8	0	1	21.0 ± 1.0	1.78 ± 0.05	70.8 ± 5.4
(DA Clark et al., 2009)	Journal	10	6	4	1	18.6	1.75	67.2
(Sedlácek et al., 2015)	Journal	57	57	0	2	14 to 31	N/D	N/D
(Van Den Tillaar & Gamble, 2017)	Conference	16	11	5	1	26.5	1.76	76.0
(Van Den Tillaar & Gamble, 2018)	Journal	16	11	5	1	26.4 ± 9.0	1.76 ± 0.08	75.5 ± 12.3
(Van Den Tillaar, 2018)	Conference	11	11	0	1	22.3 ± 5.8	1.82 ± 0.08	74.8 ± 7.6
(K Clark et al., 2021)	Journal	14	14	0	1	18.0 ± 2.5	1.76 ± 0.07	70.4 ± 6.4
(Van Den Tillaar, 2021)	Journal	24	12	12	1	22.0	1.75	67.6

N/D: no data. Standard deviation values are reported when available except for references 50 and 51 that are reported ranged.

screening and selection is shown in Figure 1, following the "Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA)" criteria (Moher et al., 2009).

Description of studies included

Table 1 shows the characteristics of the articles included and the characteristics of the sample. A total of 296 subjects were included (240 men and 56 women), aged between 14 and 31 years old; measuring between 1.73 m and 1.82 m in height; and with a body mass between 66.2 kg and 77.0 kg.

Methods Characteristics

The particularities of each study in the data collection protocol are available in Table 2. Of the 16 articles included, 10 used non-motorized TS with elastic cords or pulleys, exercised via the action of other people or counterweights, while 6 studies used different types of motorized TS.

Results of the kinematic variables

Table 3 summarizes the results of the included studies on the kinematic variables (velocity increases, stride length, stride rate, eccentric contact time, concentric contact time, contact time and flight time).

Table 2. Methods data from the analysed studies.

Reference	Towing system	Load	Data zone	Start type	Tests methods	Recovery
(Mero & Komi, 1985)	Another athlete by tightened rubber rope	30–45 N (uniform)	10-m	Flying with 35-m acceleration	6 runs (2 sub-maximal – 2 maximal – 2 supra-maximal)	5–7 min
(Bosco & Vittori, 1986)	Motorized system designed by the authors	100–150 N (uniform)	10-m	Flying with 50-m acceleration	A series of runs (not determined)	No data
(Mero & Komi, 1986)	Another athlete by tightened rubber rope	30–45 N (uniform)	10-m	Flying with 35-m acceleration	6 runs (2 sub-maximal – 2 maximal – 2 supra-maximal)	5–7 min
(Mero et al., 1987)	Motorized system (Bosco & Vittori, 1986) with rubber rope	±25 N	10-m	Flying with 35-m acceleration	10 runs (3 maximal – 5 supra-maximal – 2 maximal)	5-5-26-5-5-24-5-5-5 min
(Mero & Komi, 1987)	Another athlete by tightened rubber rope	30–45 N (uniform)	10-m	Flying with 35-m acceleration	10 runs (6 sub-maximal – 2 maximal – 2 supra-maximal)	5–7 min
(Corn & Knudson, 2003)	Another athlete by elastic band Power system Speed Harness	40–50 N	20-m	Standing	4 runs (2 maximal – 2 supra-maximal)	5 min
(Leblanc & Gervais, 2004)	Rope and pulley Stroops Double-Time overspeed trainer	No data	10-m	Flying with 30-m in Free and 20-m in Assisted	12 runs (3 free – 3 resisted – 3 assisted – 3 sprint start)	No data
(Sedlácek et al., 2004)	Towed by weights and pulley with Speedy system	20, 30, 40, 50 N	20-m	Flying with 20-m acceleration	2 maximal and 2–8 supra-maximal with different loads	No data
(Sugiura & Aoki, 2008)	Motorized system Nishi Sports	5 kg (isotonic)	6-m	Flying with 60-m acceleration	4 runs (2 maximal – 2 supra-maximal)	10–20 min
(DA Clark et al., 2009)	Towed by Theraband elastic tubing combinations	2.0, 2.8, 3.8, 4.7% body weight	60-m	Standing	5–9 runs in each 5 conditions in different days with minimum 48 hours of rest	5–7 min
(Sedlácek et al., 2015)	Towed by weights and pulley with Speedy system	20, 30, 40, 50 N	20-m	Flying with 20-m acceleration	2 maximal and 2–8 supra-maximal with different loads	No data
(Van Den Tillaar & Gamble, 2017)	Towed by cord, pulleys and weights	82 N	30-m	Standing	9 runs (3 maximal – 3 assisted – 3 resisted)	No data
(Van Den Tillaar & Gamble, 2018)	Toweb by cord, pulleys and weights	82 N	30-m	Standing	9 runs (3 maximal – 3 assisted – 3 resisted)	No data
(Van Den Tillaar, 2018)	Motorized system Dynaspeed	3, 4, 5 kg	60-m	Standing	4 runs (1 maximal – 1 in each towed condition)	No data
(K Clark et al., 2021)	Motorized system 1080 Sprint	7 kg (isotonic)	20-m	Flying with 20-m acceleration	4 runs (2 maximal – 2 supra-maximal)	6 min
(Van Den Tillaar, 2021)	Motorized system Dynaspeed	3, 4, 5 kg	60-m	Standing	4 runs (1 maximal – 1 in each towed condition)	10–12 min

Running speed ability or technical data in overspeed with towing system

Only seven studies (K Clark et al., 2021; DA Clark et al., 2009; Corn & Knudson, 2003; Leblanc & Gervais, 2004; Mero & Komi, 1986, 1987; Sugiura & Aoki, 2008) analysed how TS in OS affects technical or biomechanical factors comparing the athlete's technical model in conditions of maximum speed without OS and in OS with TS.

The horizontal distance between the start of contact and the vertical projection of the centre of gravity increased significantly in OS with TS in: from 0.27 m to 0.31 m ($p < 0.01$) in Mero et al. (Mero et al., 1987); from 0.15 m to 0.22 m ($p < 0.001$) in Corn and Knudson (Corn & Knudson, 2003), and from 0.41 m to 0.43 m ($p = 0.046$) in Clark et al. (DA Clark et al., 2009), with 4.7% body weight load. It non-significantly increased from 0.24 m to 0.26 m in Mero and Komi (Mero & Komi, 1985).

The study by Clark et al. (K Clark et al., 2021) shows how displacement of the centre of gravity during stride phases increased significantly in OS with TS, both during the contact phase (+3.7% $p < 0.001$) and flight phase (+13.1% $p < 0.001$). In terms of the angle of the hip at the point of contact, Clark et al. (DA Clark et al., 2009) found significant differences from −39º in the no towing condition to −49º ($p = 0.044$) with a TS condition with a load equivalent to 4.7% of body weight.

Significant increases in supramaximal speed in the knee angle at the start of contact were reported in Mero and Komi (Mero & Komi, 1985) from 153º to 157º ($p < 0.001$); in Mero et al. (Mero et al., 1987) from 146º to 152º ($p < 0.001$) and in Sugiura and Aoki (Sugiura & Aoki, 2008) from 154º to 161º ($p < 0.01$). On the contrary, Sugiura and Aoki (Sugiura & Aoki, 2008) described that the knee angle significantly decreased at the end of contact from 161º to 157º ($p < 0.01$).

The inclination of the ground shank was significantly reduced in OS with TS in Mero and Komi (Mero & Komi, 1985) from 89º to 87º ($p < 0.05$) and in Mero et al. (Mero et al., 1987) from 89º to 86º ($p < 0.001$). However, the angle of the shank at the beginning of contact increased significantly in the study by Sugiura and Aoki (Sugiura & Aoki, 2008) from 8º to 10º ($p < 0.05$). No other parameters analysed in these studies showed significant changes. The rest of the articles included in this review did not analyse biomechanical parameters related to the speed competition technique.

Electromyographic activity

Five studies included in this review recorded electromyographic (EMG) activity data during OS with TS (Mero & Komi, 1986, 1987; Mero et al., 1987; Sugiura & Aoki, 2008; Van Den Tillaar & Gamble, 2018). Mero and Komi (Mero & Komi, 1986) reported significant increases in supra-maximal speed in EMG activity of the vastus lateralis and gastrocnemius muscles in the Men B (PB 100-m: 10.96 ± 0.19s) group during the eccentric phase of contact ($p < 0.001$) and in the Women (PB 100-m: 12.22 ± 0.28s) group in the concentric phase ($p < 0.05$) when compared to maximal speed. Another study by the same authors (Mero & Komi, 1987) analysed pre-activation EMG activity during contact, post-activation EMG activity, and minimum EMG activity in ten muscles (gastrocnemius, biceps femoris, gluteus maximum, rectus femoris and vastus lateralis in the first run and tibialis anterioris, latissimus dorsi, rectus abdominis, biceps brachii and triceps brachii in the second run). The study described how pre-activation of the five leg muscles increased significantly ($p < 0.001$) in all conditions, from submaximal to supramaximal, when MRS and OS were compared. Post-activation values of the five leg muscles decreased non-significantly in OS when comparing with MRS. No significant changes were noted in terms of minimum activity of the five leg muscles. Furthermore, another study by Mero et al. (Mero et al., 1987) did not record changes in EMG activity in the five muscles assessed (gastrocnemius, vastus lateralis, biceps femoris, rectus femoris and gluteus maximus), neither during the contact phase nor during flight phase.

The study by Sugiura and Aoki (Sugiura & Aoki, 2008) assessed EMG activity in the biceps femoris and rectus femoris in conditions of MRS and OS. In the analysis of two consecutive strides, the rectus femoris increased its activity (121 ± 13% $p < 0.05$) in Flight 1 (from the moment one foot leaves the ground to the contact of the contralateral foot) ($p < 0.05$). However, the biceps femoris increased its activity (121 ± 23% $p < 0.01$) in Flight 2 (from the moment the second foot leaves the ground to the first foot touching the ground again) ($p < 0.01$) in supramaximal speed. The EMG activity of the biceps femoris also decreased in the contact phase in OS (88 ± 7% $p < 0.01$). Lastly, Van den Tillaar and Gamble (Van Den Tillaar & Gamble, 2018), by measuring values step by step and only found significant differences ($p < 0.006$) in the increase in EMG activity for the biceps femoris and semimembranosus muscles in strides 1 to 4.

Ground reaction forces, spring constants and metabolic data

Three studies (Mero & Komi, 1986, 1987; Mero et al., 1987) analysed ground reaction forces (GRF) values with force platforms in OS conditions. To describe and analyse GRF, these studies divided the analysis of ground contact into the eccentric phase and the concentric phase. During the concentric phase, Mero and Komi (Mero & Komi, 1986) demonstrated that the values of the average resultant force decreased non-significantly at supramaximal speed, while in another two studies (Mero & Komi, 1987; Mero et al., 1987) no changes were reported. Conversely, significant increases were noted during the eccentric phase. Mero and Komi (Mero & Komi, 1986) described how GRF increased significantly ($p < 0.05$) in two groups, Women and Men A (PB 100-m: 10.62 ± 0.04s), and non-significantly in the Men B group. Mero (Mero et al., 1987) reported significant increases ($p < 0.01$–0.001) in the GRF parameters observed. This study also analysed vertical force, reporting a significant increase ($p < 0.05$–0.01) in vertical force production, both in terms of maximum values and average values in supramaximal speed compared with maximal speed. Lastly, significant increases ($p < 0.001$) were observed in the production of the resulting average force in this phase (Mero & Komi, 1987).

The only study that analysed data on factors related to spring constants (or measures of stiffness) was the study by Mero and Komi (Mero & Komi, 1986). During the eccentric

Table 3. Kinematic factors in maximal speed compared with overspeed in towing system conditions.

Reference	Δ vel (m/s)	SL (m)	SR (step/s)	CT ecc (s)	CT con (s)	CT (s)	FT (s)
(Mero & Komi, 1985)	9.25 to 10.04 (+8.5%) $p < 0.001$	2.06 to 2.20 (+6.8%) $p < 0.001$	4.49 to 4.56 (+1.7%) ns $p < 0.10$	0.043 to 0.037 (−13.9%) ns	0.059 to 0.059 ns	0.102 to 0.096 (−5.9%) $p < 0.001$	0.121 to 0.125 (+3.3%) $p < 0.05$
(Bosco & Vittori, 1986)	10.15 to 11.60 (+14.3%) $p < 0.01$	2.26 to 2.45 (+8.4%) $p < 0.02$	4.50 to 4.70 (+4.44%) ns	N/D	N/D	0.093 to 0.085 (−8.6%) $p < 0.01$	0.130 to 0.126 (−3.1%) ns
(Mero & Komi, 1986)	9.45 to 10.26 (+8.6%) $p < 0.001$	ns Δ	Sig Δ $p < 0.05$–0.001	0.046 to 0.047 (+3.0%) ns	0.059 to 0.052 (−11.8%) ns	0.105 to 0.100 (−5.3%) ns	N/D
(Mero et al., 1987)	9.85 to 10.27 (+4.4%) $p < 0.001$	2.12 to 2.20 (+3.8%) $p < 0.01$	4.65 to 4.67 (+0.43%) ns	0.044 to 0.046 (+4.5%) ns	0.052 to 0.050 (−3.9%) ns	0.096 to 0.095 (−1.0%) ns	0.120 to 0.127 (+5.8%) $p < 0.01$)
(Mero & Komi, 1987)	9.57 to 10.36 (+8.3%) $p < 0.001$	N/D	N/D	0.046 to 0.047 (+3.0%) ns	0.059 to 0.052 (−11.8%) ns	0.105 to 0.100 (−5.3%) ns	0.124 to 0.127 (+2.4%) ns
(Corn & Knudson, 2003)	8.06 to 8.63 (+7.1%) $p < 0.001$	1.90 to 2.03 (+6.8%) $p < 0.001$	4.26 to 4.25 (−0.23%) $p < 0.445$	N/D	N/D	N/D	N/D
(Leblanc & Gervais, 2004)	9.40 to 10.04 (+6.8%) ns	2.26 to 2.28 (+0.9%) ns	4.26 to 4.51 (+5.9%) ns	N/D	N/D	0.114 to 0.103 (−9.6%) ns	N/D
(Sedlácek et al., 2004)	9.23 to 10.33 (+10.66%) $p < 0.01$	2.10 to 2.34 (+11.4%) $p < 0.01$	4.41 to 4.43 (+0.45%) ns	N/D	N/D	0.113 to 0.105 (−7.1%) $p < 0.01$	0.114 to 0.121 (+6.1%) $p < 0.01$
(Sugiura & Aoki, 2008)	9.5 to 10.22 (+7.5% ± 0.5) $p < 0.001$	2.26 to 2.34 (+3.4% ± 2.6) $p < 0.01$	4.22 to 4.38 (+4.0% ± 3.0) $p < 0.01$	N/D	N/D	0.114 to 0.107 (−6.1%) $p < 0.01$	0.124 to 0.122 (−1.6%) ns
(DA Clark et al., 2009)	(+5.4% for 3.8% BW and +6.1% for 4.7% BW) $p < 0.01$	ns ch in all conditions	(+4.3% for 3.8% BW ($p < 0.01$) and +4.0% for 4.7% BW) $p < 0.05$	N/D	N/D	Sig ch in all conditions $p < 0.01$	ns Δ in all conditions
(Sedlácek et al., 2015)	9.23 to 10.33 (+10.66%) $p < 0.01$	2.10 to 2.34 (+11.4%) $p < 0.01$	4.41 to 4.43 (+0.45%) ns	N/D	N/D	0.113 to 0.105 (−7.1%) $p < 0.01$	0.114 to 0.121 (+6.1%) $p < 0.01$)
(Van Den Tillaar & Gamble, 2017)	7.04 to 7.48 (+6.2) $p < 0.05$	(+5%) $p < 0.05$	ns ch	N/D	N/D	(−2.8%) $p < 0.05$	(+4.5%) $p < 0.05$
(Van Den Tillaar & Gamble, 2018)	7.04 to 7.48 (+6.2%) $p < 0.05$	Sig Δ $p < 0.001$	N/D	N/D	N/D	Sig decreases $p < 0.001$	Sig Δ $p < 0.001$
(Van Den Tillaar, 2018)	(+2.1% for 3 kg, +3.9% for 4 kg, +5.1% for 5 kg) $p < 0.05$	Sig Δ with 3, 4 and 5 kg $p < 0.05$	ns ch $p < 0.63$	N/D	N/D	ns ch $p < 0.06$	ns ch $p < 0.073$
(K Clark et al., 2021)	10.0 to 10.9 (+9.4%) $p < 0.001$	2.11 to 2.30 (+8.7%) $p < 0.001$	4.72 to 4.75 (+0.7%) ns $p = 0.36$	N/D	N/D	0.100 to 0.095 (−5.2%) $p < 0.001$	0.112 to 0.116 (+3.4%) $p < 0.05$
(Van Den Tillaar, 2021)	Sig Δ for male and female with 3, 4 and 5 kg $p < 0.05$	Sig Δ for male and female with 3, 4 and 5 kg $p < 0.05$	ns ch for male and female with 3, 4 and 5 kg	N/D	N/D	Sig decreases for male and female with 5 kg $p < 0.05$	ns ch for male and female with 3, 4 and 5 kg

Δ vel: velocity increases; SL: step length; SR: step rate; CT ecc: contact time of the eccentric phase; CT con: contact time of the concentric phase; CT: contact time of concentric and eccentric phase; FT: flight time; BW: body weight; ns: non-significant; Sig: significant; Δ: increases; ch: changes; N/D: no data.

phase, the values increased significantly as speed increased from submaximal to maximal but the values only significantly decreased in supramaximal speed in women ($p < 0.05$). During the concentric phase, no significant differences were observed.

Only Mero et al. (Mero et al., 1987) reported post-repetition peak blood lactate and oxygen debt data to assess the anaerobic metabolism in OS conditions. They noted how the values analysed in OS were significantly lower in comparison with maximal speed, 27.7% ($p < 0.001$) in peak blood lactate and 30.3% ($p < 0.01$) in oxygen debt.

Meta-analyses

Figure 2 shows the summary effect sizes and heterogeneity data of the meta-analyses performed for the different kinematic outcomes.

Velocity increases

When analysing the effects of overspeed conditions on Δ vel (Figure 2 and Figure S1 of supplementary materials), high levels of heterogeneity were observed amongst studies ($I^2 = 84.06\%$, $\tau^2 = 1.16$). Compared to the maximal sprint condition, Δ vel was higher in the overspeed condition with moderate to very large effect (ES: 1.54, 95% CI [0.94, 2.14], $p < 0.001$). Sub-group analysis could not be performed due to insufficient degrees of freedom.

Step length

When analysing the effects of overspeed conditions on SL (Figure 2 and Figure S2 of supplementary materials), medium levels of heterogeneity were observed amongst studies ($I^2 = 72.84\%$, $\tau^2 = 0.43$). Compared to the maximal sprint condition, SL was higher in the overspeed condition with small to large effect (ES: 0.92, 95% CI [0.57, 1.28], $p < 0.001$). Sub-group analysis could not be performed due to insufficient degrees of freedom.

Step rate

When analysing the effects of overspeed conditions on SR (Figure 2 and Figure S3 of supplementary materials), low levels of heterogeneity were observed amongst studies ($I^2 = 0.00\%$, $\tau^2 = 0.00$). Compared to the maximal sprint condition, SR was higher and non-statistically significant in the overspeed condition with trivial to small effect (ES: 0.12, 95% CI [−0.01, 0.25], $p < 0.082$).

Contact time of the eccentric phase

When analysing the effects of overspeed conditions on CT of the eccentric phase (Figure 2 and Figure S4 of supplementary materials), medium levels of heterogeneity were observed amongst studies ($I^2 = 34.87\%$, $\tau^2 = 0.33$). Compared to the maximal sprint condition, CT of the eccentric phase was lower and non-statistically significant in the overspeed condition, ranging from small and lower to small and higher effect (ES: −0.11, 95% CI [−0.50, 0.29], $p = 0.597$).

Contact time of the concentric phase

When analysing the effects of overspeed conditions on CT of the concentric phase (Figure 2 and Figure S5 of supplementary materials), low levels of heterogeneity were observed amongst studies ($I^2 = 23.57\%$, $\tau^2 = 0.28$). Compared to the maximal sprint condition, CT of the concentric phase was lower in the overspeed condition with trivial to moderate effect (ES: −0.49, 95% CI [−0.89, −0.08], $p = 0.018$).

Contact time of concentric and eccentric phase

When analysing the effects of overspeed conditions on CT of concentric and eccentric phase (Figure 2 and Figure S6 of supplementary materials), medium levels of heterogeneity were observed amongst studies ($I^2 = 34.51\%$, $\tau^2 = 0.06$). Compared to the maximal sprint condition, CT of concentric and eccentric phase was lower in the overspeed condition with small to moderate effect (ES: −0.57, 95% CI [−0.77, −0.37], $p < 0.001$).

Variable	ES [95% CI]	p	Q	p	I² (%)	τ²
Δ vel	1.54 [0.94, 2.14]	< 0.001	112.96	< 0.001	84.06	1.16
SL	0.92 [0.57, 1.28]	< 0.001	74.29	< 0.001	72.84	0.43
SR	0.12 [-0.01, 0.25]	0.082	7.34	0.987	0.00	0.00
CT ecc	-0.11 [-0.50, 0.29]	0.597	9.75	0.203	34.87	0.33
CT con	-0.49 [-0.89, -0.08]	0.018	6.86	0.444	23.57	0.28
CT	-0.57 [-0.77, -0.37]	< 0.001	17.00	0.113	34.51	0.06
FT	0.28 [0.09, 0.48]	0.004	31.16	0.039	40.03	0.07

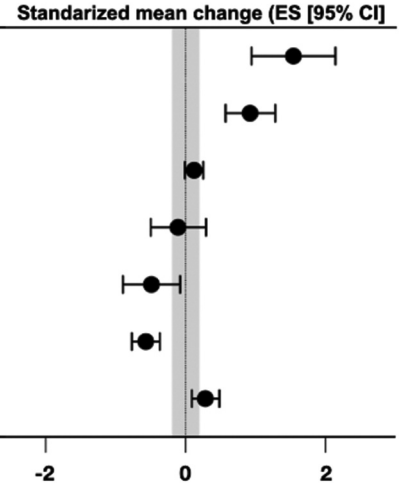

Figure 2. Effect of the overspeed condition compared to the maximal-sprint condition intervention on kinematic variables. Bars indicate uncertainty in the true mean changes with 95% confidence limits. Shaded area indicates the trivial effect size value. Δ vel: velocity increases; SL: step length; SR: step rate; CT ecc: contact time of the eccentric phase; CT con: contact time of the concentric phase; CT: contact time of concentric and eccentric phase; FT: flight time; ES: effect size; CI: confidence interval.

Flight time

When analysing the effects of overspeed conditions on FT (Figure 2 and Figure S7 of supplementary materials), medium levels of heterogeneity were observed amongst studies (I^2 = 40.03%, τ^2 = 0.07). Compared to the maximal sprint condition, FT was higher in the overspeed condition with trivial to small effect (ES: 0.28, 95% CI [0.09, 0.48], p = 0.004).

Risk of bias

The results of the risk of bias assessment are presented in Figure 3. Moderate bias was found regarding confounding factors inherent to lack of randomization, as well as the blinding of outcome assessors and participants.

Discussion

The aim of this systematic review with meta-analysis was to summarize and analyse the existing studies available among the scientific literature regarding exposure to acute effects of OS with TS in sprinters and describe how it affects different kinematic, biomechanical, and neuromuscular parameters. The different methods used to create the OS conditions were also synthesized in terms of the type of TS, the magnitude of the overload used, and data collection conditions. The results show us that, although OS with TS improves the MRS acutely of an athlete, it is not possible to accurately determine the potential mechanisms responsible for the reported changes. Mainly, it cannot be assured that this is due to a greater muscular action or to the action of the device that pulls the athlete forward.

Figure 3. Risk of bias summary by item and study. Green marker: Low risk of bias; Yellow marker: Moderate risk of bias; Red marker: Serious risk of bias. D1: Bias due to confounding; D2: Bias due to selection of participants; D3: Bias in classification of interventions; D4: Bias due to deviations from indented interventions; D5: Bias due to missing data; D6: Bias in measurement of outcomes; D7: Bias in selection of the reported result.

Furthermore, the great diversity of the TS and the lack of standardization of the methodologies and overloads used do not allow us to conclude concisely whether OS training could be useful to achieve specific long-term adaptations. In this regard, the new motorized and computerized TS can be of great help to individualize the load for each athlete and to monitor the results immediately.

Overspeed with towing system and maximal speed

In Table 3 it can be observed that TS produces speed increases in the athlete in a range that goes from + 2.1% for 3 kg (Van Den Tillaar, 2018) to + 14.3% (Bosco & Vittori, 1986). Therefore, we can state that this method allows the athlete to run at higher speeds than under unassisted conditions. However, two major questions still exist: 1) which load is the most optimal to use during a training period? 2) Will these speed increases be reproducible by the athlete without the help of the TS after the training period?

Responding to the first question, the existing literature recommends speeds over 105–110% of the MRS (DA Clark et al., 2009; Sedlácek et al., 2004, 2015) but special attention is paid to the fact that the loads used should be those that cause the least possible distortion in the kinematic and running technique parameters of the athlete's natural pattern (K Clark et al., 2021; Hicks, 2017; Leblanc & Gervais, 2004; Sedlácek et al., 2015). In five of the articles (Van Den Tillaar, 2021; DA Clark et al., 2009; Sedlácek et al., 2004, 2015; Van Den Tillaar, 2018) the effects of various loads are analysed to decide which would be the best load based on the changes they produce in kinematic and technical parameters. In the study by Clark et al. (DA Clark et al., 2009) they conclude that loads that lead to 10% increases of the MRS can be excessive. Only loads over 3.8% of body weight are recommended, producing changes of + 4.8% of the MRS without significantly altering the horizontal distance from the support to the vertical projection of the centre of gravity. This is an aspect that worsens the technical running pattern, but as Leyva et al (Leyva et al., 2017) point out, they did not carry out any study with a training period. In two more studies (Sedlácek et al., 2004, 2015) various loads are analysed with the conclusion that loads over 20–40 N may be optimal, producing MRS increases of 105–108%. Van den Tillaar recommends loads of 5 kg which represent improvements of + 4.8% of MRS (Van Den Tillaar, 2018) and in his latest work (Van Den Tillaar, 2021), the application of assisted loads of 4–5 kg depends on the sex, anthropometrics and level of the athlete.

It can be concluded that speeds close to or greater than + 10% of the MRS may be excessive and cause acute detrimental changes to sprinting mechanics. Loads that produce increases over + 5% of MRS may be optimal, and even lighter loads that allow running above the athlete's speed barrier. However, this needs to be supported through longitudinal training studies. Recommended values of 20–40 N (Sedlácek et al., 2004, 2015) of absolute load may be excessively broad for application in training, as these suggested values do not take into account sex differences or other individual differences between athletes. Van den Tillaar (Van Den Tillaar, 2018), exclusively contemplated male athletes, so their use in females cannot be generalized. However, this author's most recent study (Van Den Tillaar, 2021) provides results differentiated by sex. Among his main conclusions, he remarks that, in the 60 metres analysed, the different speed assistance loads applied produce greater time decreases in women than in men. These differences are attributed to the fact that the loads used were the same for both genders (3, 4 and 5 kg), so the relative propulsive force was greater for women (body mass: 60.7 ± 5.1 kg) than for men (body mass: 74.5 ± 6.8 kg). In future studies, it would be interesting to analyse the effects of the same relative loads.

This brings us to the second question: the lack of studies with an intervention period with different loads in sprinters. The authors are not aware of studies with longitudinal periods of overspeed training with different loads and their effects.

Therefore, more studies should be conducted, both to determine the acute effects of different loads and to see the effects of those loads after longitudinal periods of training. It is also important to be able to standardize the magnitude and the effects of the load. Possibly, the best option is to use the % of body weight (K Clark et al., 2021; DA Clark et al., 2009) to determine the relative effects of the different loads on each athlete and to be able to compare them. According to Van den Tillaar (Van Den Tillaar, 2021), monitoring the sprints step-by-step could be another method to determine the optimal load, since it would match to the load that allows to continue running for a longer time at the maximal speed achieved, without alterations in the kinematics of steps. However, the data taken consider an absolute load, so we could also consider a relative load, transforming the data into a % of body weight).

Overspeed and kinematic perspective

Specific literature has described that sprint performance is the product of SL and SR (Monte et al., 2017; Salo et al., 2011). Consequently, increasing one or both of these components without negatively impacting the other will lead to an increase in speed performance (Faccioni, 1994; Hauschildt, 2010; Mero et al., 1992; Monte et al., 2017; Salo et al., 2011). In this regard, only Sugiura and Aoki (Sugiura & Aoki, 2008) have shown how TS may significantly increase both SL and SR as specific stimuli, among other mechanical aspects related to MRS. It is important to know the characteristics of the athletes to determine if OS can be an effective training method. In this regard, Salo et al. (Salo et al., 2011) explain that the performance of a sprinter can be SL or SR reliant and that this must be taken into account when selecting training methods. The SL reliant sprinters must pay special attention to maintaining strength levels, while the SR reliant sprinters depend more on the level of activation of the neural system. Sugiura and Aoki (Sugiura & Aoki, 2008) split athletes into pitch-type (those in which the relationship between their height and stride length at MRS is less than 1.3 and are more SR-depending) and stride-type (those in which this relationship is equal or higher than 1.3 and are more SL-depending). They report that the use of TS increases SL values in pitch-type athletes, while the TS increases SR values in stride-types athletes. In Van den Tillaar's study (Van Den Tillaar, 2021) men have longer step length, mainly due to their higher body height and longer limbs, but women show greater increases than men, possibly due to their lower body mass, that allows the system to pull them with a greater relative force. Finally, Sedlácek et al. (Sedlácek et al.,

2015) argue that athletes with higher SR values can benefit more from OS training. Therefore, coaches should know the characteristics of their athletes in addition to performing sessions of familiarization with overspeed training to determine if it may be a good method for them.

After analysing the articles included in this review, it can be speculated that an increase in SL during TS stimulation is mainly owed to the inherent action of the towing mechanism (Corn & Knudson, 2003; Sedlácek et al., 2004; Sugiura & Aoki, 2008; Van Den Tillaar & Gamble, 2017, 2018) and not to the athlete's muscle action, as explained later. It is important to note that there are great differences in the methodology used between articles (See Table 2) to record and analyse data, in addition to OS with TS conditions. For this reason, it is extremely difficult to establish comparisons, and even less conclusions.

Regarding SR, the most frequent conditions caused by TS may have a direct impact on the greater neuromuscular activation of the muscles involved in the MRS, and consequently on a potential increased adaptive response capacity (Faccioni, 1994, 1994; Mero et al., 1992; Schiffer, 2011) due to the higher production of a specific force (Mero et al., 1992). Although authors (Monte et al., 2017; A Ross et al., 2001) have described how speed training may improve nerve conduction velocity, they could not demonstrate how this effect occurs in parallel to a SR improvement. According to Monte et al. (Monte et al., 2017), these relationships between SR and greater or lesser nervous activation are "athlete-dependent" and other mechanical and physiological factors, responsible for the effect on MRS, are involved. Therefore, the individual response of each athlete to acute exposure to OS over SL and SR should be analysed attending to individual characteristics before scheduling training periods in the medium or long-term.

The majority of studies included in this review reported decreases in CT with the TS method (see Table 3). Nevertheless, the division of contact time in the eccentric (braking) phase and concentric (propulsion) phase (Mero & Komi, 1985, 1986, 1987; Mero et al., 1987) show how this reduction in CT appears only during the concentric phase and even increases during the eccentric phase. These conclusions raise doubts as to whether the possible reduction in CT with TS is an adaptive neuromuscular response or if it simply occurs because the athlete ceases to prematurely push and is carried along by the system (Sedlácek et al., 2015; Sugiura & Aoki, 2008). Similarly, FT shows a tendency to increase as the speed with the TS increases (see Table 3). Some authors attributed this to the aforementioned effect of the system, rather than it being a consequence of a great application of force by the athlete (Sedlácek et al., 2015; Sugiura & Aoki, 2008). Therefore, the main interest in analysing the acute effects of OS with TS must focus on discerning what occurs in muscle action and not on variables resulting from the same, mainly SL, CT and FT.

Overspeed and ground reaction forces

Only three articles (Mero & Komi, 1986, 1987; Mero et al., 1987) presented GRF data, showing how the increase in force with TS mainly appears during the eccentric contact phase, which may mean that the athlete uses more force to brake than in propulsion (concentric phase), since the values tend to decrease during this phase. This increase in braking forces must be minimized with the correct selection of the OS load because, as Nagahara et al. (Nagahara et al., 2018) described, the achievement of high speeds is associated with a greater application of force horizontally with an anterior-posterior direction and a decrease in braking forces. However, it would be relevant to analyse data that refers to the vertical ground reaction forces, since the possibility of achieving greater speeds in the maximal speed phase depends on increasing the forces that are reached with a vertical component, as concluded in different studies (Colyer, Nagahara, Takai et al., 2018; Nagahara et al., 2018; Weyand et al., 2000). This would allow an optimal ratio to be maintained between SL and SR and also to prevent a descent in the centre of gravity, which would increase with TS stimuli (Mero et al., 1992). This would be an undesired effect caused by the TS on the athlete's mechanics and motor pattern. Despite its importance, only the article by Mero et al. (Mero et al., 1987) included an analysis of these GRF about the vertical component, with significant increases ($p < 0.05$–0.01) in both maximum values and average values during TS. In short, it would be necessary to analyse the OS with TS phenomenon from a GRF perspective using the means and methods available today, particularly force plates, which are superior in terms of precision and reliability. This has been carried out with MRS (Colyer, Nagahara, Salo et al., 2018; Colyer, Nagahara, Takai et al., 2018; Nagahara et al., 2018), and it should be possible to determine the exact influence the TS has on the athlete's speed and adjust loads individually for training periods.

Overspeed and electromyographic activity

The impact of TS on the neuromuscular system could be analysed by measuring the EMG activity of muscles specifically involved during the sprint (Howard et al., 2018; Mero et al., 1992). As described in the results, EMG activity was analysed in some articles included in this review (Mero & Komi, 1986, 1987; Mero et al., 1987; Sugiura & Aoki, 2008; Van Den Tillaar & Gamble, 2018) but it is not possible to establish relevant conclusions due to the methodological differences in their analysis, with little data reported. Despite the lack of data, a relationship is established between the GRF data in the direction of traction and the increase in the EMG activity of the muscles involved in the braking phase. Mero & Komi (Mero & Komi, 1986), Sugiura & Aoki (Sugiura & Aoki, 2008) and Van den Tillaar and Gamble (Van Den Tillaar & Gamble, 2018) also report changes in hamstring activity, especially in the biceps femoris and the semimembranosus during the leg recovery phase in the flight phase, and in the rectus femoris (Sugiura & Aoki, 2008). This combination of muscle actions can be associated with quicker leg movements with less force applied horizontally with an anterior-posterior direction. The latter is not recommended, because – as Haugen et al. (Haugen et al., 2018) described – although sprinters must quickly seek to recover the leg after propulsion to search for a new contact, this action must not be performed at the expense of reducing the full extension of the ankle. This is because greater values of this impulse are related to superior horizontal speeds and improved sprint performances. More research in OS is needed, with wireless surface EMG (Howard et al., 2018) or textile electrodes embedded into clothing (Colyer & McGuigan, 2018; Finni et al., 2007), which can be useful for the analysis of muscle actions in all the phases of sprinting, in a more ecological approach.

Overspeed and sprint technique

The technical model of speed performance has been widely studied in the literature (Colyer, Nagahara, Takai et al., 2018; Haugen et al., 2018, 2019; Mero et al., 1992; Morin et al., 2011), but only seven of the articles included in this review analysed biomechanical parameters related to the sprint technique (K Clark et al., 2021; DA Clark et al., 2009; Corn & Knudson, 2003; Leblanc & Gervais, 2004; Mero & Komi, 1985; Mero et al., 1987; Sugiura & Aoki, 2008). In this regard, OS with TS conditions should not cause great biomechanical differences or differences in the technical sprint model between a speed stimulus with or without TS (K Clark et al., 2021; Sedlácek et al., 2004, 2015), a fact that would decrease the specificity of the training stimulus (Corn & Knudson, 2003). However, we cannot state this with certainty, since there are not enough studies that analyse possible modifications in the sprint technique after a period of overspeed training, similarly to the study of Lahti et al (Lahti, Huuhka et al., 2020) after a heavy resisted sprint training period.

Three main aspects stood out: 1) The horizontal distance between the start of contact and the vertical projection of the centre of gravity increases of 14.8% ($p < 0.01$) in Mero et al. (Mero et al., 1987) and of 46.7% ($p < 0.001$) in Corn and Knudson (Corn & Knudson, 2003). On the contrary, in the study by Clark et al. (DA Clark et al., 2009), only the 5.7% increase with 4.7% of body weight acquires significance ($p < 0.046$). This is an undesired effect of the TS as it increases GRF in the eccentric phase of CT, as mentioned previously, and it must be avoided because it causes losses in horizontal speed, particularly when combined with the increase in SL (DA Clark et al., 2009). The cause is the sprinter's inertia, which continues to move forward given the system's action (Mero et al., 1992). In this vein, Clark et al. (K Clark et al., 2021) described how the centre of gravity was more displaced during FT than during CT as a result of the TS' action. 2) The angle of the knee was analysed at two points: at the beginning and at the end of contact. It can be stated that this angle in TS tends to increase significantly during the first phase of contact, as reported by Mero and Komi (Mero & Komi, 1985), Mero et al. (Mero et al., 1987) and Sugiura and Aoki (Sugiura & Aoki, 2008). In other words, the leg comes into contact with the ground with a greater knee extension. As a result, the athlete would actively and voluntarily look for contact with the ground beforehand, therefore increasing EMG activity in pre-activation (Mero & Komi, 1987) to respond to the greater GRF generated in the eccentric phase (Mero & Komi, 1986). The same angle at the end of the contact phase tends to be significantly reduced, as described by Sugiura and Aoki (Sugiura & Aoki, 2008). This effect may be attributed to the action of towing system and the athlete removing their foot prior to full propulsion in order to quickly search for the next point of contact. The study by Mero and Komi (Mero & Komi, 1987) describes how EMG activity in post-activation decreases, although not significantly. 3) The ground-shank angle recorded at the beginning of contact decreases its values significantly from 89º to 87º ($p < 0.05$) in Mero and Komi (Mero & Komi, 1985) and from 89º to 86º ($p < 0.001$) in Mero et al. (Mero et al., 1987) as a result of greater GRF in the eccentric phase. This may also be related to the decrease in spring constants (Mero & Komi, 1987) and a greater vertical oscillation of the centre of gravity (Mero et al., 1992).

Therefore, research into the acute effects of OS with TS, as well as possible adaptations to training with these systems, must focus on thoroughly analysing muscle actions via GRF and EMG activity. The methodology and towing systems should be standardized, as opposed to focusing so much on the responses or temporary adjustments that are caused on a kinematic or technical level, with the aim of recommending (or not) its use to improve MRS.

Possible intervention studies must also observe the changes that occur at other levels, such as physiological, molecular and even epigenetic changes. In this respect, only the study by Mero et al. (Mero et al., 1987) analysed what happened in metabolic terms, showing higher concentrations of lactate in the blood and higher oxygen debt in MRS than in OS, which may indicate less muscle action. Moreover, despite muscle stiffness is another physiological quality that would be expected to improve with OS stimuli (Faccioni, 1994; Monte et al., 2017), only one of the studies included in this review reported data in this regard. Reduced muscle elasticity or spring constants were described during the braking phase, with no changes in the impulse phase (Mero & Komi, 1986). This response could be explained by the increase in braking forces registered in the eccentric contact phase (Mero & Komi, 1986, 1987; Mero et al., 1987).

Future perspectives: the path to individualization in overspeed training

As is the case in other sports with multifactorial phenotypes, adaptations to speed and overspeed training are complex and influenced by both environmental and genetic factors (Murtagh et al., 2020; Rundqvist et al., 2019). Consequently, a combination of interindividual variability in genetic and epigenetic factors related to OS training could help to understand the differences between "high responders" and "low or non-responders" to overspeed training (Lahti, Jiménez-Reyes et al., 2020). Maximal speed training increases motor unit firing (Behm, 1995), leading to a higher rate of force production if better motor unit synchronization exists (Hauschildt, 2010). Thus, MRS performance improvements would be achieved if adaptive physiological responses occur mainly in muscle and neural tissues. Neuromuscular improvements caused by OS training are attributed to improvements in motor unit synchronization, due to the increased load on the muscle during the stretch-shortening cycle (Flanagan & Comyns, 2008). Future lines of research must analyse what happens at these levels (neuromuscular and molecular) to be able to determine the effects of OS training with TS, mainly in intervention studies with monitoring.

Conclusions

The main acute effects of OS with TS found in this review are the increase of the MRS, SL and FT values and the decrease of CT values. Despite the observed changes, it cannot be determined if they are due to modified neuromuscular activation and ground force application or to the device.

Knowing the acute effects of OS with TS is a first step to use it as a training method to improve MRS in the long term. Future research in this area of knowledge about sports performance should take into account the following recommendations:

(1)) The use of TS devices that allow control and the application of loads, with the aim of individualizing and selecting those that may cause improvements and may be quantified.
(2)) Comprehensively analyse muscle actions using GRF and/or EMG data to provide further insight and resolution during exposure to OS.
(3)) Being able to obtain elite sprinter data for this type of training.
(4)) The development of more intervention studies monitoring the effects of OS training with TS.

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